

第七章 复数

Chapter 7 Complex Numbers

7.1 复数的概念 Concepts of Complex Numbers

核心知识归纳 Core Knowledge Summary

1. 数系扩充与虚数单位 Number System Extension and Imaginary Unit

- 扩充背景: 为解决方程 $x^2 + 1 = 0$ (或 $x^2 + 5 = 0$) 在实数集中无解的问题, 引入虚数单位 i 。

Extension Background: To solve the problem that the equation $x^2 + 1 = 0$ (or $x^2 + 5 = 0$) has no solution in the set of real numbers, the imaginary unit i is introduced.

- 规定: $i^2 = -1$; i 可与实数进行四则运算, 运算律保持不变。

Stipulation: $i^2 = -1$; i can perform four arithmetic operations with real numbers, and the operational laws remain unchanged.

2. 复数的定义与分类 Definition and Classification of Complex Numbers

- 定义: 形如 $z = a + bi$ ($a, b \in \mathbb{R}$) 的数叫做复数, 其中:

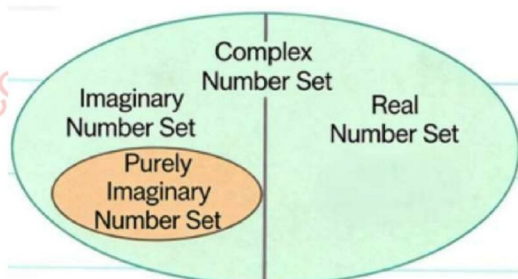
Definition: A number of the form $z = a + bi$ ($a, b \in \mathbb{R}$) is called a complex number, where:

- a 为实部 ($\operatorname{Re}(z) = a$), b 为虚部 ($\operatorname{Im}(z) = b$);

a is the real part ($\operatorname{Re}(z) = a$), and b is the imaginary part ($\operatorname{Im}(z) = b$);

- 全体复数构成复数集 $C = \{a + bi \mid a, b \in \mathbb{R}\}$, 实数集 R 是 C 的真子集 ($R \subsetneq C$)。

All complex numbers form the set of



complex numbers $C = a + bi \mid a, b \in R$, and the set of real numbers R is a proper subset of C ($R \subsetneq C$).

- 分类: Classification:

- 实数: $b = 0$ (如 3 、 $-\sqrt{2}$ 、 0);

Real Numbers: $b = 0$ (e.g., 3 , $-\sqrt{2}$, 0);

- 虚数: $b \neq 0$ (如 $2 + 3i$ 、 $-0.2i$);

Imaginary Numbers: $b \neq 0$ (e.g., $2 + 3i$, $-0.2i$);

- 纯虚数: $a = 0$ 且 $b \neq 0$ (如 i 、 $\sqrt{3}i$).

Pure Imaginary Numbers: $a = 0$ and $b \neq 0$ (e.g., i , $\sqrt{3}i$).

- 关键: 若两个复数有大小关系, 则它们必为实数; 虚数不能比较大小。

Key Point: If two complex numbers have a size relationship, they must be real numbers; imaginary numbers cannot be compared in size.

3. 复数相等的充要条件 Necessary and Sufficient Condition for Equality of Complex Numbers

- 设 $z_1 = a + bi$, $z_2 = c + di$ (a 、 b 、 c 、 $d \in R$), 则 $z_1 = z_2 \Leftrightarrow a = c$ 且 $b = d$ 。

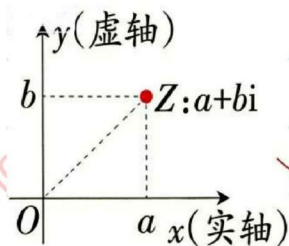
Let $z_1 = a + bi$ and $z_2 = c + di$ (a , b , c , $d \in R$), then $z_1 = z_2 \Leftrightarrow a = c$ and $b = d$.

- 前提: a 、 b 、 c 、 d 必须为实数, 否则不能直接等同实部和虚部。

Prerequisite: a , b , c , and d must be real numbers; otherwise, the real parts and imaginary parts cannot be directly equated.

4. 复数的几何意义 Geometric Meaning of Complex Numbers

- 复平面:建立直角坐标系, x 轴为实轴(表示实数), y 轴为虚轴(除原点外表示纯虚数);



For the complex number $z = a + bi$, z should be written in lowercase; for the point $Z(a, b)$ in the complex plane, Z should be written in uppercase.

复数 $z = a + bi$ 与复平面内的点

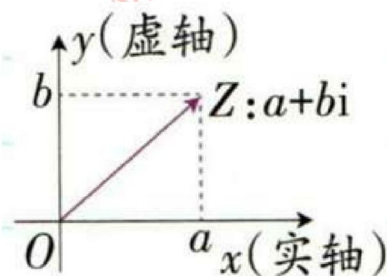
$Z(a, b)$ 一一对应(点的坐标是

(a, b) , 而非 (a, bi))。

复数 $z = a + bi$ $\longleftrightarrow$ 复平面内的点 $Z(a, b)$

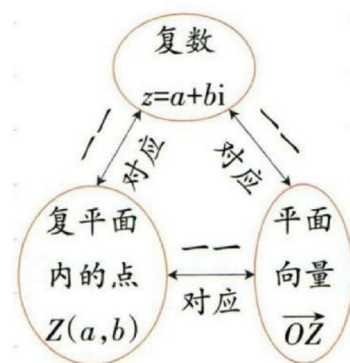
Complex Plane: Establish a rectangular coordinate system, the x -axis is the real axis (representing real numbers), and the y -axis is the imaginary axis (representing pure imaginary numbers except the origin); the complex number $z = a + bi$ is in one-to-one correspondence with the point $Z(a, b)$ in the complex plane (the coordinates of the point are (a, b) , not (a, bi)).

- 向量表示:复数 $z = a + bi$ 与复平面内以原点 O 为起点、 $Z(a, b)$ 为终点的向量 $\overrightarrow{OZ}$ 一一对应(实数 0 与零向量对应)。



复数 $z = a + bi$ $\longleftrightarrow$ 平面向量 $\overrightarrow{OZ}$

Vector Representation: The complex number $z = a + bi$ is in one-to-one correspondence with the vector $\overrightarrow{OZ}$ in the complex plane with the origin O as the starting point and $Z(a, b)$ as the endpoint (the real number 0 corresponds to the zero vector).



5. 复数的模与共轭复数 Modulus and Conjugate Complex Number of Complex Numbers

- 模(绝对值): 向量 $\overrightarrow{OZ}$ 的长度, 记作 $|z|$ 或 $|a+bi|$, 即 $|z| = \sqrt{a^2 + b^2}$; 实数的模等于其绝对值。

Modulus (Absolute Value): The length of the vector $\overrightarrow{OZ}$, denoted as $|z|$ or $|a+bi|$, i.e., $|z| = \sqrt{a^2 + b^2}$; the modulus of a real number is equal to its absolute value.

- 共轭复数:

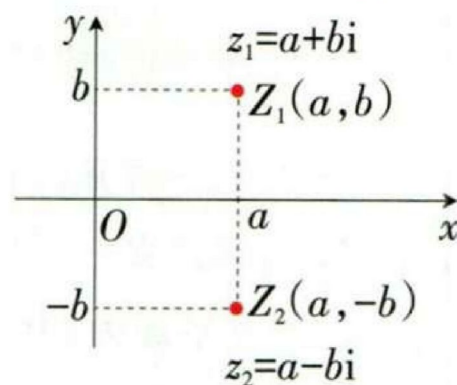
Conjugate Complex Number:

- 定义: 实部相等、虚部互为相反数的复数, 记作 $\bar{z}$, 即 $\overline{a+bi} = a-bi$; 实数的共轭复数是其本身。

Definition: A complex number with equal real parts and opposite imaginary parts, denoted as $\bar{z}$, i.e., $\overline{a+bi} = a-bi$; the conjugate complex number of a real number is itself.

- 几何意义: 复平面内对应的点关于实轴对称。

Geometric Meaning: The corresponding points in the complex plane are symmetric about the real axis.



- 性质: $\overline{(\bar{z})} = z$; $z + \bar{z} = 2a$ (实数); $z - \bar{z} = 2bi$ (纯虚数或 0); $z \cdot \bar{z} = |z|^2 = a^2 + b^2$ (实数)。

Properties: $\overline{(\bar{z})} = z$; $z + \bar{z} = 2a$ (real number); $z - \bar{z} = 2bi$ (pure imaginary number or 0); $z \cdot \bar{z} = |z|^2 = a^2 + b^2$ (real number).

易错点解析 Common Mistakes Analysis

1. 虚部的定义误区: 复数 $z = a + bi$ 的虚部是 b (实数), 而非 bi (如 $2 + 3i$ 的虚部是 3 , 不是 $3i$)。

Misunderstanding the definition of the imaginary part: The imaginary part of the complex number $z = a + bi$ is b (a real number), not bi (e.g., the imaginary part of $2 + 3i$ is 3 , not $3i$).

2. 复平面的坐标误区:虚轴的单位长度是 1 , 而非 i ; 点 $Z(a, b)$ 的纵坐标是 b , 不是 bi 。

Misunderstanding the coordinates of the complex plane: The unit length of the imaginary axis is 1 , not i ; the ordinate of the point $Z(a, b)$ is b , not bi .

3. 复数相等的条件误区:忽略 a 、 b 、 c 、 d 为实数的前提,直接等同虚部和实部。

Misunderstanding the condition for equality of complex numbers: Ignoring the prerequisite that a , b , c , and d are real numbers, and directly equating the imaginary parts and real parts.

4. 共轭复数的性质误区: $z \cdot \bar{z} = |z|^2$ 是实数,但 $|z_1 z_2| = |z_1| |z_2|$, $|z_1 + z_2| \leq |z_1| + |z_2|$ 。

Misunderstanding the properties of conjugate complex numbers: $z \cdot \bar{z} = |z|^2$ is a real number, but $|z_1 z_2| = |z_1| |z_2|$ and $|z_1 + z_2| \leq |z_1| + |z_2|$.

关键例题 Key Examples

例 1:当 m 为何值时,复数 $z = (2 + i)m^2 - 3(1 + i)m - 2(1 - i)$ 为(1)实数; (2)纯虚数?

Example 1: When does the complex number $z = (2 + i)m^2 - 3(1 + i)m - 2(1 - i)$ become (1) a real number; (2) a pure imaginary number?

解: $z = (2m^2 - 3m - 2) + (m^2 - 3m + 2)i$

Solution: $z = (2m^2 - 3m - 2) + (m^2 - 3m + 2)i$

(1) 实数: $m^2 - 3m + 2 = 0 \Rightarrow m = 1$ 或 $m = 2$;

(2) Real number: $m^2 - 3m + 2 = 0 \Rightarrow m = 1$ or $m = 2$;

(3) 纯虚数: $\begin{cases} 2m^2 - 3m - 2 = 0 \\ m^2 - 3m + 2 \neq 0 \end{cases} \Rightarrow m = -\frac{1}{2}$ 。

(4) Pure imaginary number: $\begin{cases} 2m^2 - 3m - 2 = 0 \\ m^2 - 3m + 2 \neq 0 \end{cases} \Rightarrow m = -\frac{1}{2}$.

例 2(高考真题): 设 $z = -3 + 2i$, 则 $\bar{z}$ 在复平面内对应的点位于()

Example 2 (College Entrance Examination Question): Let $z = -3 + 2i$, the point corresponding to $\bar{z}$ in the complex plane is located in ()

A. 第一象限 B. 第二象限 C. 第三象限 D. 第四象限

A. First Quadrant B. Second Quadrant C. Third Quadrant D. Fourth Quadrant

解: $\bar{z} = -3 - 2i$, 对应点 $(-3, -2)$, 位于第三象限, 选 C。

Solution: $\bar{z} = -3 - 2i$, corresponding to the point $(-3, -2)$, which is located in the third quadrant, choose C.

例 3: 已知 x 是实数, y 是纯虚数, 且 $(2x - 1) + (3 - y)i = y - i$, 求 x 、 y 。

Example 3: Given that x is a real number and y is a pure imaginary number, and $(2x - 1) + (3 - y)i = y - i$, find x and y .

解: 设 $y = bi$ ($b \in \mathbb{R}$, $b \neq 0$), 代入得 $(2x - 1 + b) + 3i = (b - 1)i$, 故 $\begin{cases} 2x - 1 + b = 0 \\ b - 1 = 3 \end{cases} \Rightarrow x = -\frac{3}{2}, y = 4i$ 。

Solution: Let $y = bi$ ($b \in \mathbb{R}$, $b \neq 0$), substitute into the equation to get $(2x - 1 + b) + 3i = (b - 1)i$

7.2 复数的四则运算

Four Arithmetic Operations of Complex Numbers

7.2.1 复数的加、减运算及其几何意义

Addition and Subtraction of Complex Numbers and Their Geometric Meanings

核心知识归纳 Core Knowledge Summary

1. 运算法则 Operation Rules

- 加法: $z_1 + z_2 = (a + c) + (b + d)i$ ($z_1 = a + bi$, $z_2 = c + di$), 实部相加, 虚部分别相加。

Addition: $z_1 + z_2 = (a + c) + (b + d)i$ ($z_1 = a + bi$, $z_2 = c + di$), where real parts are added and imaginary parts are added respectively.

$(a+bi)+(c+di)=(a+c)+(b+d)i$

Addition of complex numbers can be regarded as combining like terms with "i" treated as an unknown. The addition rule for complex numbers can be extended to the case of adding multiple complex numbers.

- 减法: $z_1 - z_2 = (a - c) + (b - d)i$, 减法是加法的逆运算, 实部相减, 虚部分别相减。

Subtraction: $z_1 - z_2 = (a - c) + (b - d)i$, subtraction is the inverse operation of addition, where real parts are subtracted and imaginary parts are subtracted respectively.

$(a+bi)-(c+di)=(a-c)+(b-d)i$

Subtraction of complex numbers can be regarded as combining like terms with "i" treated as an unknown.

- 本质: 将 i 看作“未知数”, 合并同类项。

Essence: Treat i as an "unknown quantity" and combine like terms.

2. 运算律 Operational Laws

- 交换律: $z_1 + z_2 = z_2 + z_1$;

Commutative Law: $z_1 + z_2 = z_2 + z_1$;

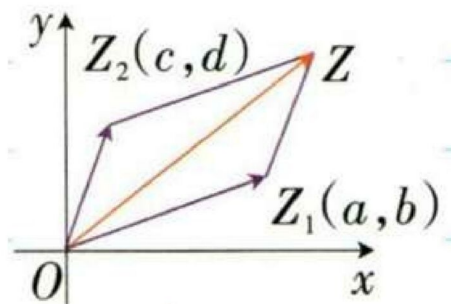
- 结合律: $(z_1 + z_2) + z_3 = z_1 + (z_2 + z_3)$ 。

Associative Law: $(z_1 + z_2) + z_3 = z_1 + (z_2 + z_3)$.

3. 几何意义 Geometric Meanings

- 加法: 对应向量的平行四边形法则(同起点)或三角形法则(首尾相接), 与平面向量加法一致。

Addition: Corresponding to the parallelogram rule (same starting point) or triangle rule (head-to-tail connection) of vectors, consistent with the addition of plane vectors.



- 减法: 对应向量 $\overrightarrow{OZ_1} - \overrightarrow{OZ_2} = (a - c, b - d)$, 即 $\overrightarrow{Z_2Z_1}$ (从 z_2 的对应点指向 z_1 的对应点)。

Subtraction: Corresponding to the vector $\overrightarrow{OZ_1} - \overrightarrow{OZ_2} = (a - c, b - d)$, i.e., $\overrightarrow{Z_2Z_1}$ (from the corresponding point of z_2 to the corresponding point of z_1).

- 距离公式: 复平面内两点 $Z_1(a, b)$ 、 $Z_2(c, d)$ 的距离 $|Z_1Z_2| = |z_1 - z_2| = \sqrt{(a - c)^2 + (b - d)^2}$ 。

Distance Formula: The distance between two points $Z_1(a, b)$ and $Z_2(c, d)$ in the complex plane is $|Z_1Z_2| = |z_1 - z_2| = \sqrt{(a - c)^2 + (b - d)^2}$.

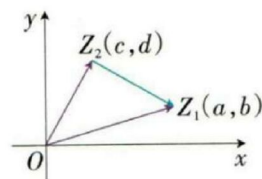
易错点解析 Common Mistakes Analysis

1. 虚部符号失误: 减法运算中, 虚部是 $b - d$, 而非 $d - b$ (如 $(2 + 3i) - (1 - 2i) = 1 + 5i$)。

Mistake in the sign of the imaginary part: In subtraction, the imaginary part is $b - d$, not $d - b$ (e.g., $(2 + 3i) - (1 - 2i) = 1 + 5i$).

2. 距离公式误区: $|z_1 - z_2|$ 表示两点间距离, 而非 $|z_1| - |z_2|$ 。

Misunderstanding the distance formula: $|z_1 - z_2|$ represents the distance between two points, not $|z_1| - |z_2|$.



3. 结果判断误区: 两个虚数的差不一定是虚数(如 $(1 + 3i) - 3i = 1$)。

Misjudgment of the result: The difference between two imaginary numbers is not necessarily an imaginary number (e.g., $(1 + 3i) - 3i = 1$).

关键例题 Key Examples

例 1(高考真题): 设复数 z 满足 $|z - i| = 1$, z 在复平面内对应的点为 (x, y) , 则()

Example 1 (College Entrance Examination Question): Let the complex number z satisfy

$|z - i| = 1$, and the point corresponding to z in the complex plane is (x, y) , then ()

A. $(x + 1)^2 + y^2 = 1$ B. $(x - 1)^2 + y^2 = 1$ C. $x^2 + (y - 1)^2 = 1$ D. $x^2 + (y + 1)^2 = 1$

解: $z = x + yi$, $|z - i| = |x + (y - 1)i| = \sqrt{x^2 + (y - 1)^2}$, 故 $x^2 + (y - 1)^2 = 1$, 选 C。

Solution: $z = x + yi$, $|z - i| = |x + (y - 1)i| = \sqrt{x^2 + (y - 1)^2}$, so $x^2 + (y - 1)^2 = 1$, choose C.

例 2: 计算 $(5 - 6i) + (-2 - i) - (3 + 4i) = -11i$ 。

Example 2: Calculate $(5 - 6i) + (-2 - i) - (3 + 4i) = -11i$.

7.2.2 复数的乘、除运算 Multiplication and Division of Complex Numbers

核心知识归纳 Core Knowledge Summary

1. 乘法法则: $z_1 z_2 = (a + bi)(c + di) = (ac - bd) + (ad + bc)i$ 。

Multiplication Rule: $z_1 z_2 = (a + bi)(c + di) = (ac - bd) + (ad + bc)i$.

- 本质: 多项式相乘, $i^2 = -1$, 合并实部和虚部。

Essence: Multiply like polynomials, substitute $i^2 = -1$, and combine real and

imaginary parts.

- 运算律:交换律 $z_1 z_2 = z_2 z_1$; 结合律 $(z_1 z_2) z_3 = z_1 (z_2 z_3)$; 分配律 $z_1 (z_2 + z_3) = z_1 z_2 + z_1 z_3$ 。

Operational Laws: Commutative Law $z_1 z_2 = z_2 z_1$; Associative Law $(z_1 z_2) z_3 = z_1 (z_2 z_3)$; Distributive Law $z_1 (z_2 + z_3) = z_1 z_2 + z_1 z_3$.

- 常用公式(平方差、完全平方):

Common Formulas (Difference of Squares, Perfect Squares):

- $(a + bi)(a - bi) = a^2 + b^2$ (共轭复数相乘为实部平方加虚部平方);

$(a + bi)(a - bi) = a^2 + b^2$ (the product of conjugate complex numbers is the sum of the square of the real part and the square of the imaginary part);

- $(a \pm bi)^2 = a^2 - b^2 \pm 2abi$ 。

2. 除法法则:除法是乘法的逆运算, $z_1 \div z_2 = \frac{a+bi}{c+di} = \frac{(a+bi)(c-di)}{c^2+d^2}$ ($c + di \neq 0$)。

Division Rule: Division is the inverse operation of multiplication, $z_1 \div z_2 = \frac{a+bi}{c+di} = \frac{(a+bi)(c-di)}{c^2+d^2}$ ($c + di \neq 0$).

- 关键:分子分母同乘分母的共轭复数 $c - di$, 实现分母“实数化”(类似根式除法的分母有理化)。

Key: Multiply both the numerator and denominator by the conjugate complex number $c - di$ of the denominator to “rationalize the denominator” (similar to rationalizing the denominator in radical division).

3. 常用结论 Common Conclusions

- i 的幂的周期性: $i^1 = i$, $i^2 = -1$, $i^3 = -i$, $i^4 = 1$, $i^{4n+k} = i^k$ ($n \in N^*$, $k = 1, 2, 3$); $i^n + i^{n+1} + i^{n+2} + i^{n+3} = 0$ 。

Periodicity of Powers of i : $i^1 = i$, $i^2 = -1$, $i^3 = -i$, $i^4 = 1$, $i^{4n+k} = i^k$ ($n \in \mathbb{N}^*$, $k = 1,2,3$); $i^n + i^{n+1} + i^{n+2} + i^{n+3} = 0$.

- 特殊复数 $\omega = -\frac{1}{2} + \frac{\sqrt{3}}{2}i$ 的性质: $\omega^2 = -\frac{1}{2} - \frac{\sqrt{3}}{2}i$, $\omega^3 = 1$, $1 + \omega + \omega^2 = 0$, $\bar{\omega} = \omega^2$.

Properties of the Special Complex Number $\omega = -\frac{1}{2} + \frac{\sqrt{3}}{2}i$: $\omega^2 = -\frac{1}{2} - \frac{\sqrt{3}}{2}i$, $\omega^3 = 1$, $1 + \omega + \omega^2 = 0$, $\bar{\omega} = \omega^2$.

4. 复数范围内的一元二次方程 Univariate Quadratic Equations in the Complex Number Field

- 实系数方程 $ax^2 + bx + c = 0$ ($a \neq 0$):

Real-Coefficient Equation $ax^2 + bx + c = 0$ ($a \neq 0$):

- $\Delta = b^2 - 4ac \geq 0$:实根 $x = \frac{-b \pm \sqrt{\Delta}}{2a}$,

- $\Delta = b^2 - 4ac \geq 0$: Real roots $x = \frac{-b \pm \sqrt{\Delta}}{2a}$;

- $\Delta < 0$:虚根 $x = \frac{-b \pm \sqrt{-\Delta}}{2a}i$ (共轭虚根)。

- $\Delta < 0$: Imaginary roots $x = \frac{-b \pm \sqrt{-\Delta}}{2a}i$ (conjugate imaginary roots).

- 性质:实系数方程的虚根成对出现; 复系数方程无此性质。

Property: The imaginary roots of real-coefficient equations appear in pairs; this property does not hold for complex-coefficient equations.

易错点解析 Common Mistakes Analysis

1. 乘法运算中 i^2 失误:忘记将 i^2 化为 -1 (如 $(1+i)(2+i) = 1 + 3i$,而非 $2 + 3i$)。

Mistake in i^2 during multiplication: Forgetting to substitute $i^2 = -1$ (e.g., $(1+i)(2+i) = 1 + 3i$, not $2 + 3i$).

2. 除法分母实数化误区:分母乘自身共轭复数,而非分子的共轭复数。

Misunderstanding of rationalizing the denominator in division: Multiply the denominator by its own conjugate complex number, not the conjugate complex number of the numerator.

3. 幂的周期性误用:忽略 i 的幂周期为 4 ,直接硬算高次幂。

Misuse of the periodicity of powers: Ignoring that the period of powers of i is 4 and directly calculating high powers.

4. 复系数方程误区:将实系数方程的虚根成对性质推广到复系数方程。

Misunderstanding of complex-coefficient equations: Extending the property of conjugate imaginary roots of real-coefficient equations to complex-coefficient equations.

关键例题 Key Examples

例 1(高考真题):已知 $z = 2 + i$,则 $z \cdot \bar{z} = ()$

Example 1 (College Entrance Examination Question): Given $z = 2 + i$, then $z \cdot \bar{z} = ()$

A. $\sqrt{3}$ B. $\sqrt{5}$ C. 3 D. 5

解: $z \cdot \bar{z} = |z|^2 = 2^2 + 1^2 = 5$,选 D。

Solution: $z \cdot \bar{z} = |z|^2 = 2^2 + 1^2 = 5$, choose D.

例 2(高考真题):若 $z(1 + i) = 2i$,则 $z = ()$

Example 2 (College Entrance Examination Question): If $z(1 + i) = 2i$, then $z = ()$

A. $-1 - i$ B. $-1 + i$ C. $1 - i$ D. $1 + i$

解: $z = \frac{2i}{1+i} = \frac{2i(1-i)}{(1+i)(1-i)} = \frac{2+2i}{2} = 1 + i$,选 D。

Solution: $z = \frac{2i}{1+i} = \frac{2i(1-i)}{(1+i)(1-i)} = \frac{2+2i}{2} = 1 + i$, choose D.

例 3:解方程 $x^2 + x + 1 = 0$ (复数范围内)。

Example 3: Solve the equation $x^2 + x + 1 = 0$ (in the complex number field).

解: $\Delta = 1 - 4 = -3 < 0$, 根为 $x = \frac{-1 \pm \sqrt{3}i}{2}$ 。

Solution: $\Delta = 1 - 4 = -3 < 0$, the roots are $x = \frac{-1 \pm \sqrt{3}i}{2}$.